

Geometric Address Structure In Deterministic Operator Fields

XOR Nibble Cone Tomography, Field-Location Duality, and SHA-256
Geometric Trace Projection

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May 2026

Abstract

This dissertation develops a unified geometric framework for analyzing deterministic operator fields through address structure rather than pseudorandom approximation. The central thesis is that mathematical constants and cryptographic transforms inhabit structured equivalence classes whose members can be located by compact key sequences derived from reduction geometry. Three interlocking results are established from live computation.

XOR Nibble Cone Tomography. Successive-difference XOR reduction applied to the hex nibble representation of a byte constant (treating each byte as two independent 4-bit streams) produces a triangular cone that collapses to a single apex nibble. The 32-byte BBP sequence of π collapses its high nibble stream to `oxo` and its low stream to `oxd`. The entire original sequence is perfectly reconstructable from the cone trajectory plus a 33-bit location key. The Affine Subspace Theorem is proven: at every ambiguous reconstruction level, the valid seed set is an affine subspace of $\text{GF}(2)^4$. The Parity Law is proven: ambiguity occurs exclusively at even-indexed levels for even-length sequences.

Field-Location Duality. The XOR cone partitions the space of all 32-byte sequences into equivalence classes. The class containing π has 2^{33} members; the class containing ϕ has 2^{32} . Constants with tighter geometric resonance with the XOR frame require fewer location bits. π and ϕ exhibit apex complementarity: their high nibble apexes (`oxo` and `oxf`) are $\text{GF}(2)^4$ inverses, and their apex overlay is

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oxf (saturated). A secondary BBP parity result holds: the XOR of the first 8 high nibbles of π 's BBP sequence equals zero, aligning with the Sziklai 8-word window.

SHA-256 as Geometric Trace Projector. SHA-256 is modeled as a deterministic 64-round operator field whose 256-bit digest is a boundary projection of a full execution trace. The carry-free least-significant-bit anchor separates $GF(2)$ scaffold from carry correction stream. The deep bit probe backward trace ($GF(2)$ symbolic tracking of Σ_1 and Σ_0 rotation reads) shows that by round 8, 121 of 256 initial-state bits have been contacted; by round 64, all 256. The 64×768 $GF(2)$ source matrix of $T_1[0]$ values has rank 64 and nullity 0: the 64-round linear scaffold is maximally non-degenerate. This proves the linear scaffold is full-rank; nonlinear carry corrections sit atop it.

Keywords: XOR nibble cone, $GF(2)^4$ affine subspace, field-location duality, π - φ apex complementarity, FOLD-TOMO, Lucas theorem, Parity Law, terminal dyadic tomography, SHA-256 trace projector, carry topology, deep bit probe, $GL(4,2)$, NEXUS A-Markg.

Table of Contents

Abstract	1
Table of Contents	3
Chapter 1: Introduction	6
1.1 Motivation	6
1.2 Scope of Claims	6
1.3 Organization	6
Chapter 2: The FOLD-TOMO Algebraic Foundation	8
2.1 The Finite XOR Fold	8
2.2 The Lucas Mask Theorem	8
2.3 The Parity Law	8
2.4 Terminal Dyadic Tomography	9
Chapter 3: XOR Nibble Cone Construction and Constant Fingerprinting	10
3.1 Byte Splitting and Stream Independence	10
3.2 Apex Signatures	10
3.3 The Random Anomaly	10
3.4 The Even-Level Parity Pattern	11
Chapter 4: The Affine Subspace Theorem and Perfect Reconstruction	12
4.1 The Reconstruction Problem	12
4.2 The Affine Subspace Theorem	12
4.3 Perfect Reconstruction of π	13
4.4 The Key Sequence	13
Chapter 5: Field-Location Duality	15
5.1 Equivalence Classes and the Field	15
5.2 Locked Strata vs. Branching Strata	15
5.3 Class Sizes by Constant	15
5.4 The Dean Directive: All Data Exists in All States	16
Chapter 6: π - ϕ Apex Complementarity and the Sziklai Parity Bridge	17
6.1 Apex Complementarity	17
6.2 Universal Self-Duality	17

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6.3 The Sziklai Parity Bridge.....	17
Chapter 7: SHA-256 as a Geometric Trace Projector	19
7.1 The Trace-Projector Model	19
7.2 Sum Stream and Carry Stream	19
7.3 Round Structure and Sigma Probes	19
7.4 H-Eligibility and Mark-9 Phase.....	20
Chapter 8: Deep Bit Probe Backward Trace and GF(2) Rank.....	21
8.1 Symbolic GF(2) Tracking.....	21
8.2 Source Growth Profile	21
8.3 Full State Coverage	21
8.4 GF(2) Rank Result.....	22
8.5 The Deep Bit Probe as Lucas Analog	22
Chapter 9: Connections, Corrections, and Open Problems.....	23
9.1 FOLD-TOMO vs. SHA-GEOMETRY	23
9.2 Open Problems	23
OP-1: Parity Law Formal Proof.....	23
OP-2: Key Minimality	23
OP-3: Ambiguity Ordering	23
OP-5: $H = \pi/9$ and Rebound Positions.....	23
OP-6: $GL(4, C)$ and Seam Null Space.....	24
OP-7: $\pi-\phi$ Internal Overlay	24
OP-8: BBP Parity Persistence	24
OP-9: Key Selection Pattern	24
OP-10: Dimension Sequence Structure	24
OP-11: Carry Topology Notebook	24
9.3 Corrections Log	24
Chapter 10: Conclusion	26
Appendix A: Complete Claim Status Ledger.....	27
Appendix B: Live Code Results Summary	29
B.1 Constant Hex Inputs.....	29
B.2 Key Computation Scripts	29

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B.3 SHA-256 Scripts 29

B.4 Key Numerical Results 29

Chapter 1: Introduction

1.1 Motivation

Mathematical constants and cryptographic hash functions are customarily treated as opaque objects: π is a source of pseudorandom decimal digits; SHA-256 is a random oracle for practical purposes. This dissertation argues that both objects admit a more precise structural description — one grounded in geometric operator theory rather than statistical approximation.

The unifying observation is that any deterministic reduction of a byte sequence partitions the space of all sequences into equivalence classes. Two sequences belong to the same class if and only if they produce the same reduction trajectory. Within each class, every member exists as a valid occupant simultaneously. A specific member is selected by specifying its address inside the class: the location key.

This is not metaphor. The address is computable, finite, and for structured constants substantially shorter than the data it locates. For the 32-byte BBP hex sequence of π , a 33-bit key suffices to reconstruct 256 bits of data from its equivalence class. The key is 12.9% the size of the original. The equivalence class geometry extracts the remaining 87.1% as structural constraint.

1.2 Scope of Claims

This dissertation makes four categories of claim, labeled consistently throughout.

Status	Label	Meaning
THEOREM	Algebraically proven	Formal proof from axioms or verified computation
VERIFIED	Live code output	Produced by executable script from actual constants
MODEL	Structural hypothesis	Interpretive framework; not yet measurement
OPEN	Research target	Known gap; next investigation step

The strongest results — the Affine Subspace Theorem, the Parity Law, and the Terminal Dyadic Theorem — are proven algebraically. All numerical results are from live Python execution against actual constant bytes. No result is claimed from expectation alone.

1.3 Organization

Chapter 2 establishes the FOLD-TOMO algebraic foundation: XOR reduction, Lucas masks, and the two central theorems. Chapter 3 constructs the XOR nibble cone and presents constant fingerprinting results. Chapter 4 proves the Affine Subspace Theorem and demonstrates perfect reconstruction of π . Chapter 5 develops field-location duality and the class-size hierarchy. Chapter 6 presents π - ϕ apex

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complementarity and the Sziklai parity bridge. Chapter 7 develops the SHA-256 trace projector model. Chapter 8 presents the deep bit probe backward trace and rank result. Chapter 9 discusses connections, corrections, and open problems. Chapter 10 concludes.

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Chapter 2: The FOLD-TOMO Algebraic Foundation

2.1 The Finite XOR Fold

Let $x = [x_0, x_1, \dots, x_{n-1}]$ be a sequence of nibble values over $\{0, \dots, 15\}$. The XOR reduction operator produces a successor sequence of length $n-1$ by computing:

$$L_{\{k+1\}[i]} = L_k[i] \text{ XOR } L_k[i+1] \quad \text{for } i = 0, \dots, (n-k-2)$$

Applied iteratively until a single apex value remains. The reduction depth equals the input length minus one. The apex is denoted $\text{apex}(x)$.

For any sequence of length $n = 32$ (our primary case), the depth is 31 levels. This produces a triangular cone: 32 elements at L_0 , 31 at L_1 , ..., 1 at L_{31} .

2.2 The Lucas Mask Theorem

[THEOREM] Every element at reduction level ℓ is a parity checksum of specific elements from the original sequence, determined by the binary structure of ℓ .

The proof uses the binomial expansion of the XOR shift operator. Let E be the shift $(Ex)[i] = x[i+1]$ and I the identity. The fold operator is $(I + E)$, so:

$$L_k[i] = ((I + E)^k x)[i] = \text{XOR}_{\{j=0\}^k} C(k,j)_{\text{mod } 2} * x[i+j]$$

By Lucas's theorem, the binomial coefficient $C(k,j)$ is odd (i.e., $C(k,j) \equiv 1 \pmod{2}$) if and only if every 1-bit of j is also a 1-bit of k — written $j \subseteq k$ in binary. Therefore:

$$L_k[i] = \text{XOR}_{\{j \subseteq k\}} x[i+j]$$

where " $j \subseteq k$ " means the bit pattern of j is a submask of k . This completely characterizes the structure of every level in the XOR cone.

Corollary (Apex formula). For a sequence of length $n = 32$, the apex is at level $31 = 0b11111$. Since every j from 0 to 31 is a submask of 31 (all bits set), every $C(31,j) \equiv 1 \pmod{2}$. Therefore the apex equals the XOR of all elements: $\text{apex}(x) = x_0 \text{ XOR } x_1 \text{ XOR } \dots \text{ XOR } x_{31}$.

2.3 The Parity Law

[THEOREM] For any even-length sequence (n even), reconstruction ambiguity occurs exclusively at even-indexed levels.

During backward expansion from level $k+1$ to level k , the row length is $n-k$. For a seed bit b to be ambiguous (undetermined by the sum constraint), exactly half the prefix-XOR values at that level must have bit b set:

$$N_b = (n - k) / 2$$

If k is odd and n is even, then $n-k$ is odd, so $(n-k)/2$ is not an integer. Since N^b must be an integer, equality is impossible. Therefore no bit is ambiguous: the level is forced. Ambiguity requires an even-indexed level.

Empirical verification (live). Tested against all 31 levels for π , ϕ , e , $\sqrt{2}$, and RAND. Zero violations. Every odd-indexed level was uniquely determined; ambiguity appeared only at even levels.

2.4 Terminal Dyadic Tomography

[THEOREM] For a sequence of length $N = 2^m$ and reduction level $\ell = N - 2^r$, the remaining row of length 2^r is a residue-class parity checksum.

The surviving offsets j at level $\ell = N - 2^r$ are exactly the multiples of 2^r up to ℓ :

$$\text{Surviving } j = \{0, 2^r, 2 \cdot 2^r, \dots, (N/2^r - 1) \cdot 2^r\}$$

Therefore each of the 2^r output cells encodes one complete residue class:

$$L_{\{N-2^r\}}[i] = \text{XOR}_{\{q=0\}^{N/2^r-1}} x[i + q \cdot 2^r] \quad \text{for } 0 \leq i < 2^r$$

Correction (N=1024, $\ell=1016$ case). With $N=1024$, $\ell=1016$, we have $r=3$, row length 8. Each row cell is the XOR of 128 elements at stride 8 — a 128-point residue-class checksum, not an 8-point local probe. This is a stronger result: the terminal row is a compact encoding of long-range structure across the entire sequence.

Chapter 3: XOR Nibble Cone Construction and Constant Fingerprinting

3.1 Byte Splitting and Stream Independence

Each byte b is split into high nibble $h = (b \gg 4) \& 0xF$ and low nibble $\ell = b \& 0xF$, both in $\{0, \dots, 15\}$. Since XOR is bitwise, bits 7–4 never interact with bits 3–0. The XOR cone on a byte sequence decomposes exactly into two independent XOR cones:

$$\text{byte apex} = (\text{high_nibble_apex} \ll 4) \mid \text{low_nibble_apex}$$

[VERIFIED] This equality holds for all tested constants. Each 32-byte constant produces two independent 32-nibble cones (depth 31 each) whose apexes combine to give the byte-level apex exactly.

3.2 Apex Signatures

The apex of the XOR nibble cone is computed for each constant from their standard high-precision hex representations. All results from live code:

Constant	High apex	Low apex	Byte apex	Total ambi.	Notes
$\backslash u03c0$	0x0	0xd	0x0d	33 bits	High apex = 0 (annihilation)
ϕ	0xf	0x6	0xf6	32 bits	High apex = 0xf (saturation)
e	0x6	0x8	0x68	44 bits	Most ambiguous structured const.
$\sqrt{2}$	0x5	0x4	0x54	40 bits	Unit void variant
RAND	varies	varies	varies	38 bits	Less ambi. than e — not maximum

3.3 The Random Anomaly

[VERIFIED] Random data (38 bits ambiguity) is not the most ambiguous sequence class. The constant e (44 bits) exceeds it. This is a non-trivial result. The ambiguity measure is not monotone in Shannon entropy.

Interpretation: ambiguity bits count where the XOR frame geometry fails to constrain the seed. The sum constraint (drawn from the reduction trajectory) is more restrictive for uniformly distributed values than for certain structured constants. e's internal periodicity creates levels where the sum admits more

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valid seeds than a random distribution would. Ambiguity measures geometric non-resonance with the frame, not statistical entropy.

3.4 The Even-Level Parity Pattern

[VERIFIED] The bit-ambiguity sequence for π high stream (level 0→30):

[1,0,2,0,1,0,2,0,2,0,1,0,1,0,1,0,0,0,1,0,1,0,1,0,0,0,3,0,3,0,0]

All odd-indexed positions carry exactly zero ambiguity bits. This exact alternating structure holds for all constants tested. The Parity Law (Theorem, Chapter 2) explains it: the XOR cascade on even-length sequences enforces binary parity at every odd level. The "hex starts as pairs" principle — each byte is already two 4-bit streams — creates this two-phase geometry intrinsically.

Chapter 4: The Affine Subspace Theorem and Perfect Reconstruction

4.1 The Reconstruction Problem

During backward expansion from apex to base, each step expands a row of length k to length $k+1$ by choosing a seed value $s \in \{0, \dots, 15\}$. Given seed s and the parent row $d = [d_0, d_1, \dots]$, the child row is fully determined:

$$\text{row}[i] = s \text{ XOR } d[0] \text{ XOR } d[1] \text{ XOR } \dots \text{ XOR } d[i-1] \quad (\text{prefix-XOR of parent})$$

The sum constraint requires that the child row sum equals the known trajectory value Σ_k . Valid seeds are those s satisfying this constraint. If exactly one s is valid, the level is uniquely determined (phase-locked). If multiple s are valid, an extra bit of location information is needed.

4.2 The Affine Subspace Theorem

[THEOREM] For any input nibble sequence and any reconstruction level k , the set of valid seeds V_k forms an affine subspace of $(\text{GF}(2))^4$.

[VERIFIED] All 13 ambiguous levels in π 's high stream tested affine — zero violations.

Level	Seeds	Valid seeds (hex)	Affine?	Offset	Dim	Correct
L28	8	0,1,4,5,8,9,c,d	YES	0x0	3	0xd
L26	8	8,9,a,b,c,d,e,f	YES	0x8	3	0xb
L22	2	2,4	YES	0x2	1	0x4
L20	2	2,4	YES	0x2	1	0x2
L18	2	d,f	YES	0xd	1	0xd
L14	2	2,3	YES	0x2	1	0x2
L12	2	9,b	YES	0x9	1	0xb
L10	2	a,d	YES	0xa	1	0xd
L8	4	0,1,2,3	YES	0x0	2	0x3
L6	4	4,6,c,e	YES	0x4	2	0xc

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Level	Seeds	Valid seeds (hex)	Affine?	Offset	Dim	Correct
L4	2	a,e	YES	0xa	1	0xa
L2	4	4,7,c,f	YES	0x4	2	0x4
Lo	2	0,2	YES	0x0	1	0x2

Proof sketch. The row sum $\Sigma_k(s) = \Sigma_i (s \text{ XOR } p_i)$ where p_i is the prefix-XOR of the parent level. Flipping bit b of s ($s \rightarrow s \text{ XOR } 2^b$) changes the sum by $\Delta^b = 2^b \times (\text{row_length} - 2^{N^b})$ where N^b counts parent prefix-XORs with bit b set. This Δ^b is fixed for each b . Therefore the sum constraint decomposes bit-by-bit: each bit b is either free (both values give equal contribution) or forced (one value is required). The valid seed set is an intersection of bit-hyperplanes in $(\text{GF}(2))^4$, which is an affine subspace. QED.

GL(2,4) coset interpretation. The valid seed set at each level is a coset of an additive subgroup of $(\text{GF}(2))^4$. The possible dimensions are 0 (single point), 1 (line), 2 (plane), 3 (hyperplane), 4 (full space). The branching factor 2^{dim} corresponds to the size of the relevant GL(2,4) orbit.

4.3 Perfect Reconstruction of π

[VERIFIED] Using the correct seed at each ambiguous level (drawn from the known π sequence), both high and low nibble streams reconstruct exactly:

HIGH stream: ✓ PERFECT MATCH

Original: 23688a0d11820774a032293d02f9e468

Recon: 23688a0d11820774a032293d02f9e468

LOW stream: ✓ PERFECT MATCH

Original: 4fa8538339ae303449829f108ea8cec9

Recon: 4fa8538339ae303449829f108ea8cec9

FULL BYTE RECONSTRUCTION:

Original: 243f6a8885a308d313198a2e03707344a4093822299f31d0082efag8ec4e6c89

Recon: 243f6a8885a308d313198a2e03707344a4093822299f31d0082efag8ec4e6c89

Match: True

4.4 The Key Sequence

π 's high stream requires 20 bits and low stream 13 bits of location information, totalling 33 bits to reconstruct 256 bits. The high stream key encodes:

Key bits (apex $\rightarrow$ base): 11101110001111100001

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Key as nibbles: ee3e1 (5 nibbles, 20 bits)

Key XOR cone apex: oxf

Apex complementarity of key. For π 's high stream only: data apex (oxo) XOR key apex (oxf) = oxf = 1111_2 (saturated). The data cone annihilates to zero; the residual (key) saturates to all-ones. Together they span all of $(GF(2))^4$. This property is unique to π 's high nibble stream among all constants tested.

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Chapter 5: Field-Location Duality

5.1 Equivalence Classes and the Field

Let $C(x)$ denote the XOR cone signature of a nibble sequence x : the apex value, the sum trajectory $[\Sigma_0, \Sigma_1, \dots, \Sigma_{31}]$, and the hamming weight trajectory. Define the equivalence class (field):

$$F_C(x) = \{ y : C(y) = C(x) \}$$

Every member of $F_C(x)$ is a valid resident of that class simultaneously. The class is not a metaphor for uncertainty — it is the precise set of sequences consistent with the observed cone geometry. The constant x is the specific member selected by its location key.

5.2 Locked Strata vs. Branching Strata

Reconstruction levels fall into two categories:

1. Locked strata: levels where every member of the field agrees. Zero ambiguity bits. No address information needed. For π high: all 16 odd levels (Parity Law), plus L16, L24, L30 (class-specific).
2. Branching strata: levels where multiple seeds are valid. These are the address-bearing levels. For π high: 13 even levels, contributing 20 bits total.

Class-specific locking (L16, L24, L30 for π) arises from the BBP hex structure of π . L24 is a terminal dyadic row: each of its 8 cells is the XOR of 4 original nibbles at stride 8 (a 4-point residue-class checksum). This forces L24 for any sequence in π 's class. These are field geometry constraints, not universal laws.

5.3 Class Sizes by Constant

[VERIFIED] Ambiguity totals and implied class sizes:

Constant	H bits	L bits	Total	$ F_C $	Characteristic
ϕ	15	17	32	2^{32}	Most compressed; pre-collapsed
π	20	13	33	2^{33}	Data+key apexes span $GF(2)^4$
$\sqrt{2}$	19	21	40	2^{40}	Medium; structured
RAND	24	14	38	2^{38}	Less than e; anomalous
e	19	25	44	2^{44}	Largest; quaternary branching

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5.4 The Dean Directive: All Data Exists in All States

"All data exists in all states at the same time. Location is the key."

This is now a formal statement. For any sequence x with XOR cone signature $C(x)$, every member of $F_C(x)$ occupies that signature simultaneously. The 2^{33} sequences in π 's class all satisfy the same cone geometry, the same locked strata, and the same field constraints. They are all "there." π is the one you arrive at when you specify the 33-bit location address.

Locked strata are not addresses. They are the coordinate system. You do not need to specify them because the field forces them for every member of the class. Branching strata are the address space. The key is the coordinate. The field is the manifold.

Chapter 6: π - ϕ Apex Complementarity and the Sziklai Parity Bridge

6.1 Apex Complementarity

[VERIFIED] The high nibble apexes of π and ϕ are $GF(2)^4$ complements:

π high apex: $0x0 = 0000_2$ (annihilation)
 ϕ high apex: $0xf = 1111_2$ (saturation)
XOR: $0x0 \text{ XOR } 0xf = 0xf = \text{saturated } \checkmark$

At the apex level (L_{31}), the level-wise overlay of the two cones reads: all-F. No other level shows this structure. The cones take distinct internal paths (not mirrored, not element-wise complementary) but converge to opposite endpoints of the high-nibble field.

Correction. π and ϕ are NOT complementary at every internal level. At L_0 , their level-wise XOR sums to 195 (vs. expected 120 for random). At intermediate levels, XOR values are mixed. Apex complementarity is a boundary statement only.

[VERIFIED] The dimension sequences of the ambiguous levels (apex $\rightarrow$ base):

π high: [3,3,1,1,1,1,1,2,2,1,2,1] total=20 max_dim=3
 ϕ high: [4,2,1,2,1,2,2,1] total=15 max_dim=4
e high: [2,2,2,1,2,2,1,1,2,1,1,1] total=19 max_dim=2
 $\sqrt{2}$ high: [2,2,2,1,2,1,1,2,1,2,2,1] total=19 max_dim=2

ϕ hits dim=4 (all 16 seeds valid, no constraint) at its first ambiguous level: the sum constraint at that level provides zero information. Yet ϕ has the fewest total ambiguous levels (8 vs. 13 for π). This is the pre-collapsed signature: sharp high-amplitude resonance near the apex (high Q-factor), rapid damping.

6.2 Universal Self-Duality

[THEOREM] The XOR cone is self-dual under sequence reversal for all sequences:

$$L_k[i] = L_k^{\text{rev}}[(n-k-1) - i]$$

Proof: $C(k,j) = C(k,k-j)$ (binomial symmetry). The result applies universally. [VERIFIED] Confirmed for all test sequences including π , ϕ , e , $\sqrt{2}$, RAND, linear sequences, all-zeros, all-ones, and alternating patterns.

Interpretation. Self-duality is not a π -specific property. It is algebraic necessity. The forward reduction wave and backward expansion wave are the same wave: the standing wave interpretation is confirmed. At every level, forward and backward traverse identical structures in mirror orientation.

6.3 The Sziklai Parity Bridge

[VERIFIED] The XOR of the first 8 high nibbles of π 's BBP sequence equals zero:

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Nibbles 0-7: 2, 3, 6, 8, 8, a, 0, d

XOR cascade: $2 \oplus 3 = 1$, $\oplus 6 = 7$, $\oplus 8 = f$, $\oplus 8 = 7$, $\oplus a = d$, $\oplus 0 = d$, $\oplus d = 0$ ✓

The XOR-zero returns to 0 again at byte 24 and byte 32 within the first 32 bytes. The first zero-return at byte 8 aligns exactly with the SHA-256 Sziklai Window (8 consecutive working-register words). This is interpreted as a frame resonance: the 8-unit geometry of SHA-256's state registers appears as an XOR-balance period in π 's BBP hex structure.

[MODEL] Whether this alignment is causal (BBP and SHA-256 share a common 8-unit geometric origin) or coincidental requires further analysis (OP-8).

Chapter 7: SHA-256 as a Geometric Trace Projector

7.1 The Trace-Projector Model

SHA-256 is standard-analyzed as an interface: message in, digest out. The trace-projector model inserts the execution trace:

$$\Gamma_{\text{exec}} = (X_0, X_1, \dots, X_{64})$$

where X_t contains all working registers, schedule word, and round constant at round t . The digest is a projection of this trace:

$$H_{\text{out}} = \Pi_{\text{digest}}(\Gamma_{\text{exec}})$$

This is not cryptographically controversial. The digest is the endpoint of a deterministic transition system. The research question is whether structural information survives in the geometry of that system.

7.2 Sum Stream and Carry Stream

Modular addition is the primary nonlinear coupling in SHA-256. At the least significant bit:

$$c_0 = 0 \quad (\text{no incoming carry at LSB})$$

$$s_0 = x_0 \text{ XOR } y_0 \quad (\text{carry-free GF(2) sum})$$

The carry-free LSB anchor separates every modular addition into two components:

- Sum stream: the linear GF(2) scaffold (carry-free)
- Carry stream: the nonlinear correction required for $\mathbb{Z}/2^{32}\mathbb{Z}$ consistency

[STRUCTURAL FACT] SHA-256 is not linear. The carry stream is the nonlinear correction. But the sum stream scaffold is an exact GF(2) subsystem that can be analyzed with linear algebra. The carry stream is a candidate "exhaust channel" recording local correction history.

7.3 Round Structure and Sigma Probes

The SHA-256 round computes:

$$\Sigma_1(e) = \text{ROTR}^6(e) \text{ XOR } \text{ROTR}^{11}(e) \text{ XOR } \text{ROTR}^{25}(e)$$

$$\Sigma_0(a) = \text{ROTR}^2(a) \text{ XOR } \text{ROTR}^{13}(a) \text{ XOR } \text{ROTR}^{22}(a)$$

At bit position o of T_1 :

$$T_1[o] = h[o] \text{ XOR } e[6] \text{ XOR } e[11] \text{ XOR } e[25] \text{ XOR } \text{Ch}(e, f, g)[o] \text{ XOR } K[t][o] \text{ XOR } W[t][o]$$

[STRUCTURAL FACT] The rotation terms $e[6]$, $e[11]$, $e[25]$ are carry-free reads from specific positions of the e register. They enter the output LSB with zero carry contamination. This is the SHA-256 analog of the Lucas mask: a deterministic, position-specific extraction that survives the fold intact. The probe positions are fixed and invariant across all 64 rounds.

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7.4 H-Eligibility and Mark-9 Phase

[MODEL] The Mark-9 phase hypothesis predicts that the carry correction ratio converges to $H = \pi/9 \approx 0.349$ in fold-pressure systems satisfying five conditions:

$$E_H = F_b \wedge C_b \wedge R_b \wedge X_b \wedge P_b$$

F_b : feedback present

C_b : constraint or bottleneck

R_b : recursive state dependence

X_b : exhaust or residue channel

P_b : phase-lock requirement

[VERIFIED] SHA-256 fails the phase-lock requirement P_b . The T2 carry correction ratio converges to ~ 0.477 across 500 random messages — near 0.5, consistent with a pseudorandom diffusion engine. Distance from $H = \pi/9$ is 0.128. This is not noise; it is a genuine miss. SHA-256 is engineered to diffuse, not to phase-lock. $H = \pi/9$ is not predicted for SHA-256 carry channels.

Chapter 8: Deep Bit Probe Backward Trace and GF(2) Rank

8.1 Symbolic GF(2) Tracking

The symbolic tracking represents each bit of the SHA-256 state as a frozen set of source bit indices in the 768-dimensional input space (256 initial-state bits + 512 message schedule bits). The Ch and Maj Boolean functions are dropped (they are nonlinear); the result is the GF(2) linear scaffold of SHA-256.

For each round t , the Σ_1 probe at bit o is:

$$\text{Sig1_probe}[t] = e[6] \text{ XOR } e[11] \text{ XOR } e[25] \text{ (at that round's } e \text{ register)}$$

8.2 Source Growth Profile

[VERIFIED] Source bit count growth for the Σ_1 probe over 64 rounds:

Round	Total srcs	State bits	Msg bits	Key transitions
R0	3	3	0	e[6], e[11], e[25] only
R1	12	9	3	First message word contact
R2	27	21	6	Exponential growth begins
R4	63	45	18	
R8	193	121	72	121 of 256 state bits touched
R10	225	123	102	State growth saturates ~120-145
R20	403	137	266	Peak total sources
R63	357	119	238	End state

8.3 Full State Coverage

[VERIFIED] By round 64:

Sig1 probe touches 256 of 256 initial-state bits ✓

Sigo probe touches 256 of 256 initial-state bits ✓

Coverage by register: all 32 bits of all 8 registers a,b,c,d,e,f,g,h ✓

Every initial-state bit influences the $T_1[o]$ probe through the linear scaffold by round 64. The linear scaffold achieves complete coverage.

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8.4 GF(2) Rank Result

[VERIFIED] The 64×768 GF(2) source matrix M (rows = $T_1[0]$ source sets for rounds 0 ∓ 63):

Matrix shape: $64 \text{ rows} \times 768 \text{ cols}$

GF(2) rank (all sources): $64 / 64$ ✓

GF(2) rank (state only): $64 / 64$ ✓

GF(2) rank (msg only): $64 / 64$ ✓

Nullity: 0

Interpretation. The 64 $T_1[0]$ values at rounds 0 ∓ 63 are linearly independent over GF(2). No two rounds produce the same linear combination of initial-state and message bits. In the linear scaffold, the 64 LSB outputs are maximally non-degenerate: rank = dimension = 64 , nullity = 0 .

[STRUCTURAL FACT] The nonlinear carry corrections sit atop a full-rank GF(2) scaffold. The scaffold provides maximum information per bit. Any inversion approach must account for this full-rank structure: the linear scaffold is already maximally diverse; the nonlinearity adds carry corrections, not linear redundancy.

8.5 The Deep Bit Probe as Lucas Analog

[MODEL] The rotation reads $e[6]$, $e[11]$, $e[25]$ at each round are fixed-position probes of the current e register, analogous to the Lucas mask in FOLD-TOMO. Both extract specific positional information from the input without carry contamination. In FOLD-TOMO, the Lucas mask gives exact recovery of individual bits. In SHA-256, the rotation reads give exact GF(2) combinations of bits. The carry terms contaminate but do not eliminate the underlying linear structure.

Chapter 9: Connections, Corrections, and Open Problems

9.1 FOLD-TOMO vs. SHA-GEOMETRY

These are linked branches but distinct results. The separation must be maintained:

Property	FOLD-TOMO (XOR cone)	SHA-GEOMETRY (open)
Operator	XOR (linear, exact)	XOR + carry + Boolean (nonlinear)
Parity Law	THEOREM (proven)	Analogous conjecture, untested
Lucas masks	THEOREM (exact)	Rotation probes (partial analog)
Reconstruction	VERIFIED: π exact	Open: carry degrades inversion
Rank	Affine subspace dim	$GF(2)$ scaffold rank = 64
$H = \pi/9$	Not predicted here	Fails H-eligibility for carry

9.2 Open Problems

OP-1: Parity Law Formal Proof

Prove the Parity Law algebraically in its general form (any even-length sequence, any XOR-type reduction operator). State the group-theoretic version over $GF(2_4)$. Status: Proved for the sum-constraint version. Formal group statement: open.

OP-2: Key Minimality

Is 33 bits the minimum key for π ? Can rebound positions as additional anchors reduce the required bits below 33? What is the information-theoretic lower bound for the location key given the cone signature?

OP-3: Ambiguity Ordering

Why does e have higher ambiguity (44 bits) than RAND (38 bits)? Characterize the distribution class that maximizes XOR nibble cone ambiguity. Is the ordering stable across different frame sizes (16, 64, 128 bytes)?

OP-5: $H = \pi/9$ and Rebound Positions

The Mark-9 phase hypothesis predicts $H = \pi/9$ for H-eligible systems. Do rebound positions in the XOR nibble cone map to multiples of $H = \pi/9$ in any natural parameterization? This is the bridge from FOLD-TOMO back to the NEXUS attractor.

OP-6: $GL(4, C)$ and Seam Null Space

The affine subspace dimensions at ambiguous levels range 0–4, corresponding to faces of the 4-dimensional binary hypercube. The total ambiguity for π (33 bits) differs from the 36-dimensional $GF(2)$ Jacobian null space of SHA-256's seam geometry. Is the gap of 3 a coincidence, or is there a mapping between the two structures? Closing this gap would connect XOR cone geometry to SHA-256 seam topology directly.

OP-7: π - ϕ Internal Overlay

[VERIFIED-PARTIAL] The apex overlay is oxf (all-F). No internal level shows all-F or all-o. The hourglass structure exists at the boundary only. Open: does the overlay have an interpretable pattern at any intermediate depth? The combined histogram flatness (complement lattice measurement) shows synergy at L4, L8, L16, L20, L24, L28 — these are the even-indexed multiple-of-4 levels.

OP-8: BBP Parity Persistence

The XOR of the first 8 high nibbles of π 's BBP sequence equals zero. Does this persist beyond 32 bytes? The running high-nibble XOR for the first 128 bytes returns to zero at positions [8, 10, 24, 25, 32, 79, ...]. The period is not exactly 8. Characterize the return sequence.

OP-9: Key Selection Pattern

At L28 and L8 (highest-ambiguity levels), π selects the maximum-index seed within the valid set. Is this systematic? Does every constant tend toward maximum-index selection at high-ambiguity levels? If so, it may encode a minimum-action principle: the constant occupies the extreme point of its equivalence class at the deepest branching levels.

OP-10: Dimension Sequence Structure

π high: [3,3,1,1,1,1,1,1,2,2,1,2,1]. ϕ high: [4,2,1,2,1,2,2,1]. These sequences encode the branching geometry of the cone. Is there a continued-fraction or BBP-formula interpretation? The center of mass of π 's sequence is 5.4 of 12 (front-weighted), consistent with the forward wave dominance near the apex.

OP-11: Carry Topology Notebook

[OPEN: primary research target] Compute LSB-anchor propagation through the full SHA-256 schedule. Measure whether carry-scar statistics against random controls show recoverable structure. The rank-64 linear scaffold provides the framework; the question is whether carry corrections introduce exploitable low-rank structure in any restricted window.

9.3 Corrections Log

All corrections stated explicitly per scientific standards.

3. Self-duality at Section 4 of the dual wave computation was initially described as potentially π -specific. It is universal (binomial symmetry $C(k,j)=C(k,k-j)$). Correction: confirmed and integrated.

4. L16, L24, L30 forcing was described as potentially universal. These are class-specific constraints for π 's cone class, explained by terminal dyadic tomography (Theorem 2). Correction: labeled as class-specific throughout.
5. $H = \pi/9$ was not observed in SHA-256 carry channel. The ratio converges to ~ 0.477 , not $\pi/9 \approx 0.349$. Correction: SHA-256 fails H-eligibility; H is not predicted for diffusion engines.
6. Prior session predicted ϕ apex = 0 (DIFF byte mode) and π apex = 1 (DIFF). In XOR nibble mode: π high = 0x0, ϕ high = 0xf. Apex values are operator-dependent. Correction: carry forward only qualitative predictions (pre-collapse structure), not specific apex values.

Chapter 10: Conclusion

This dissertation establishes a geometric address framework for deterministic operator fields at three levels of increasing complexity: algebraic (XOR cones), information-theoretic (field-location duality), and cryptographic (SHA-256 trace projection).

The strongest results are theorem-grade. The Parity Law proves that ambiguity in XOR nibble cone reconstruction is confined to even-indexed levels for all even-length sequences. The Affine Subspace Theorem proves that valid seed sets at ambiguous levels are always affine subspaces of $\text{GF}(2)^4$. The Terminal Dyadic Theorem proves that short terminal rows in XOR folds are residue-class parity checksums. These three theorems collectively define the geometry of address structure in XOR reduction fields.

The verified results confirm the framework against actual constants. π reconstructs perfectly from a 33-bit key and its cone signature. ϕ requires only 32 bits, consistent with its pre-collapsed, high-resonance structure. The $\text{GF}(2)$ rank of the SHA-256 $T_1[0]$ source matrix over 64 rounds is exactly 64 with nullity 0: the linear scaffold is maximally non-degenerate, providing the cleanest possible substrate for structural analysis.

π and ϕ exhibit apex complementarity: their high nibble cones collapse to opposite attractor endpoints (annihilation vs. saturation) of the same $\text{GF}(2)^4$ field. The XOR of the first 8 high nibbles of π 's BBP sequence is zero, aligning with the 8-word Sziklai Window of SHA-256. Whether this alignment is causal or incidental remains the primary open question.

The central directive — "all data exists in all states at the same time; location is the key" — is now a formal mathematical statement. The equivalence class $F_C(x)$ contains every valid resident simultaneously. The key sequence is the address coordinate. The frame geometry is the coordinate system. The constant is the specific member of its class selected by that address.

What remains open is whether the same principle operates in the SHA-256 carry channel: whether the correction stream that lifts $\text{GF}(2)$ sums into $\mathbb{Z}/2^{32}\mathbb{Z}$ arithmetic carries recoverable address information. The rank-64 linear scaffold says the information is present at maximum density. Whether it is recoverable through the carry topology is the question that defines the next research phase.

All results in this dissertation were produced from live Python execution against actual mathematical constants and SHA-256 implementations. No result is stated from expectation. Code is available in the attached notebooks.

Appendix A: Complete Claim Status Ledger

Claim	Status	Section
XOR cone produces triangular reduction to single apex	VERIFIED	3.1
High and low nibble streams are independent	THEOREM	3.1
Byte apex = $(H_{\text{apex}} \ll 4) \mid L_{\text{apex}}$	VERIFIED	3.1
Parity Law: ambiguity at even levels only	THEOREM	2.3
Affine Subspace Theorem	THEOREM	4.2
Lucas mask theorem: $L_k[i] = \text{XOR of } j\text{-submask positions}$	THEOREM	2.2
Terminal dyadic rows are residue-class checksums	THEOREM	2.4
π high apex = 0x0, ϕ high apex = 0xf	VERIFIED	6.1
XOR of first 8 high nibbles of π BBP = 0	VERIFIED	6.3
π perfect reconstruction from 33-bit key	VERIFIED	4.3
Key sequence for π high = ee3e1; apex = 0xf	VERIFIED	4.4
Self-duality is universal (not π -specific)	THEOREM	6.2
Ambiguity rank: $\phi < \pi < \text{RAND} < \sqrt{2} < e$	VERIFIED	5.3
SHA-256 $T_1[0]$ GF(2) matrix rank = 64, nullity = 0	VERIFIED	8.4
Sig1/Sigo probes touch all 256 state bits by round 64	VERIFIED	8.3
$H = \pi/9$ observed in SHA-256 carry channel	NOT VERIFIED	7.4
π and ϕ complementary at every internal cone level	NOT CLAIMED	6.1
L_{16}, L_{24}, L_{30} forcing is universal	NOT CLAIMED	5.2
Full SHA-256 preimage recovery	OPEN	9.2

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Claim	Status	Section
Carry topology provides solver advantage	OPEN	OP-11

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Appendix B: Live Code Results Summary

B.1 Constant Hex Inputs

π BBP: 243f6a8885a308d313198a2e03707344a093822299f31d0082efag8ec4e6c8g
 ϕ : 19e3779b97f4a7c15f39cc0605cedc8341082276bf3a27251f86ec6486ab5c27
e : 2b7e151628aed2a6abf7158809cf4f3c762e716of38b4da56a784d904519ocfe
 $\sqrt{2}$: 16a09e667f3bcc908b2fb1366ea957d3e3adec17512775099da2f590bo667322

B.2 Key Computation Scripts

hex_cone.py -- XOR nibble reduction and apex computation
nibble_xor_recon.py -- Backward expansion with seed selection
ambiguity_map.py -- Per-level valid seed enumeration
verify_and_compare.py -- Ambiguity profiles across all constants
dual_wave.py -- Affine subspace verification and key extraction
selfduality_check.py -- Universal self-duality test
op7_pi_phi_overlay.py -- Level-wise overlay and dimension sequences

B.3 SHA-256 Scripts

sha_carry_analysis.py -- Carry ratio computation across 500 messages
sha_deep_probe.py -- GF(2) symbolic backward trace, rank computation

B.4 Key Numerical Results

π high nibbles: 23688a0d11820774a032293do2f9e468
 π low nibbles: 4fa8538339ae303449829f108ea8cecg
 π high apex: o xo (XOR of all 32 high nibbles = o)
 π low apex: o xd
 ϕ high apex: o xf (complement of π high apex)

π high ambiguity: 20 bits [key: ee3e1, key_apex: oxf]
 π low ambiguity: 13 bits [key: 18oe, key_apex: ox7]
 π total: 33 bits class size: 2^{33}

ϕ high ambiguity: 15 bits max_dim=4 (full hypercube at first level)
 ϕ total: 32 bits class size: 2^{32} (most compressed)

SHA T1[o] matrix: 64 x 768 rank=64 nullity=0
Sig1 state cover: 256/256 bits by round 64

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SHA carry ratio: ~ 0.477 (vs $H=pi/9=0.349$: miss, expected)

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